Venkata Lakshmi Parimala Pasupuleti

Fraud & Risk Analytics · Income Fraud Detection · SQL · Python · Tableau · Databricks · Anomaly Detection · Compliance

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PROFESSIONAL SUMMARY

Fraud domain expert and Software Engineer with 8+ years building **income fraud detection systems** and production data pipelines at Entrata (PropTech, 100K+ MAU). Detected and blocked fake paystubs, synthetic rent histories, and AMI income misrepresentation across LIHTC/HCV/HUD portfolios — zero compliance violations. Applied **Isolation Forest anomaly detection** across multi-source research pipelines, surfacing behavioral patterns that manual review missed. Skilled in **SQL, Python, Tableau, PySpark/Databricks, and statistical analysis** with hands-on experience building KPI dashboards (precision, recall, fraud rate metrics) and cross-functional policy recommendations. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

CORE COMPETENCIES

Fraud Prevention & Detection | Income Fraud Analytics | Identity Theft & Account Takeover | Anomaly Detection (Isolation Forest) | SQL & Data Profiling | Python (scikit-learn, Pandas, NumPy) | Tableau & Data Visualization | Databricks / PySpark | KPI Metrics (Precision, Recall, Fraud Rate) | Statistical Analysis & Hypothesis Testing | Risk Analytics & Policy Recommendations | Cross-Functional Collaboration | Product Abuse & Behavioral Patterns | Customer Trust & Safety

EXPERIENCE

George Mason University — Research Associate / Teaching Assistant

Aug 2024 – May 2026

Research Associate · Anomaly Detection · Statistical Modeling · Data Visualization

- **Applied Isolation Forest anomaly detection** across 3 research pipelines — surfaced anomalies and behavioral patterns that manual review missed, improving downstream model reliability and data integrity for faculty stakeholders.
- Achieved $R^2 = 0.977$ regression accuracy across 190+ countries by building multi-source prediction models (Python, R) with rigorous **statistical analysis**, hypothesis testing, and validation pipelines on WHO and UK Road Safety datasets.
- Designed **Tableau and ggplot2 leadership dashboards** tracking KPI metrics across two cohorts — improved research comprehension by 40%; data profiling and exploratory analysis drove research strategy recommendations for faculty.
- Delivered technical workshops to 600+ students across MIS 303: Python, data structures, SQL, ER modeling, data validation, and pivot table analysis.

Magizhs Tech Solution

May 2025 – Aug 2025

AI/ML Engineer Intern — Risk Analytics · Compliance Automation · Policy Enforcement

- Built an agentic compliance pipeline (LangGraph stateful graph) applying **risk analytics and policy recommendations**: compliant operations route to export, non-compliant ones route to rejection with grounded, cited reasons — enforcing rule-based decision boundaries analogous to fraud policy engines.
- Implemented RAG-based knowledge retrieval over FAA Part 107 corpus — every compliance check functions as a **hypothesis test grounded in policy documentation**; engineered guardrails blocking non-compliant actions before execution.

Entrata (PropTech / Real Estate SaaS)

Sep 2018 – Aug 2024

Senior Software Engineer — Income Fraud Detection · Compliance Automation · Financial Data Pipelines

- **Detected and blocked income fraud** (fake paystubs, synthetic rent histories, household-composition fraud) across LIHTC, Section 8/HCV, and HUD portfolios — automated AMI threshold calculations, eligibility verification, and recertification workflows with **zero compliance violations** across 100K+ unit portfolios.
- **Built fraud analytics and risk detection systems** for affordable housing financial services: analyzed AMI threshold distributions across 1.6M HUD/Tax Credit income records (5 years, 5 US states) to detect income misrepresentation patterns; built occupancy risk dashboards tracking recertification compliance rates, waitlist anomalies, and subsidy exposure across 100K+ unit portfolios; collaborated cross-functionally with compliance, operations, and product stakeholders on policy recommendations.
- **Optimized 200+ SQL queries** (30% performance gain) through execution plan tuning, indexing, database partitioning, and audit tables — built **data profiling** tools and log tables for end-to-end traceability of financial transactions across all client environments.
- **Achieved 100% PHI compliance** on AWS S3 migration with Libsodium-encrypted pipelines (encrypted 100% of resident data — phone numbers and SSNs); zero security incidents; updated 50%+ of product codebase for encryption compliance.
- Eliminated 80% of manual operational workload via **Python automation pipelines**; built internal debugging tools reducing debugging time 40%; shipped 20+ full-stack production modules across 120+ Agile sprints for 100K+ MAU at 99.5% uptime; **mentored 5+ junior analysts and engineers**, conducted 300+ peer code reviews.
- Led **microservices architecture** redesign using Docker — 50% improvement in system scalability, 30% reduction in deployment time; processed 1.6M HUD/Tax Credit income records across 5 years via generic SQL import scripts.

Entrata India (Db Xento Systems)

Jan 2017 – Sep 2018

Associate / Trainee Software Engineer — Compliance Data Pipelines · AWS S3 · Document Management

- Built full-stack affordable housing compliance modules (PHP, PostgreSQL, JavaScript); designed SQL import scripts processing 1.6M HUD/Tax Credit income records; migrated thousands of client documents to AWS S3 and redesigned document storage database architecture; created 56 state-specific certification documents with merge fields for housing eligibility verification.

PORTFOLIO PROJECTS

CrashClarity — Large-Scale Risk Analytics & Predictive Modeling (104K Records)

Statistical Analysis | Anomaly Detection | Data Visualization

2025 · Python, R, scikit-learn, Random Forest, Logistic Regression, Feature Engineering, Tableau, GitHub Pages · View Project

- Achieved **85%+ accuracy** (precision and recall tracked across model iterations) on 104K+ UK accident records by building a **Random Forest + Logistic Regression** pipeline with feature engineering across behavioral, environmental, and temporal variables — classical ML statistical modeling applied to real-world risk prediction.
- Performed **data profiling, exploratory statistical analysis**, and hypothesis testing to identify behavioral patterns driving risk — deployed interactive dashboard on GitHub Pages for cross-functional stakeholder communication.

Leasel — Deterministic Fraud-Prevention Audit Pipeline for LIHTC Compliance

Fraud Analytics | Compliance | Policy Enforcement

2026 · Python, LangGraph (8-agent), FastAPI, TF-IDF, OCR, SQLite · github.com/venkatalakshmiparimala/Lease_complaince_rag

- Built a deterministic audit pipeline that prevents AI from making unilateral compliance decisions — LLM reasons and extracts; human logic or deterministic code makes every regulated eligibility, compliance, and financial call; **93.6% precision** on 1,240 residential lease compliance documents (5 US states + federal, 1,050 synthetic tenants).
- Implemented 8-agent LangGraph investigation graph (Triage → Retrieval → Config → Historical Incident → Structured Data → Reasoning → Validation → Human Escalation) — correctly traces **false-ineligibility to misconfigured income verification settings**; 87.9% F1 across 10 compliance rules; 100% citation correctness, 0% unsupported-claim rate.

AgentProbe — 5-Metric Evaluation Engine (Precision, Recall, Hallucination, Completeness, Compliance)

KPI Metrics | Fraud Analytics

2024–Present · Python, LangGraph, FastAPI, ChromaDB, Anthropic, OpenAI, Groq, Ollama, React, Docker, GitHub Actions · github.com/venkatalakshmiparimala/AgentProbe

- Built a **5-metric evaluation engine** measuring precision, recall, hallucination rate, completeness, and compliance — the same KPI framework used in fraud detection systems; CI-integrated via GitHub Actions with regressions blocked before merge.
- 15-endpoint FastAPI backend with 3-agent LangGraph runtime (Orchestrator, RAG Tracer, Auto Fix) and multi-provider LLM routing; trace-level observability dashboard tracking per-run metrics and drift across concurrent test suites.

EDUCATION

M.S. Data Analytics Engineering — George Mason University, Virginia

May 2026 · GPA 3.87

Coursework: Statistical Analysis, Data Mining, Machine Learning, Data Visualization, Database Systems, Applied Analytics

B.Tech Computer Science & Engineering — JNTUA, India

2013–2017

TECHNICAL SKILLS

Fraud Analytics & Risk: Fraud Prevention, Fraud Detection, Fraud Analyst, Risk Analyst, Risk Management, Income Fraud, Identity Theft Detection, Account Takeover, Product Abuse, Anomaly Detection (Isolation Forest), Behavioral Pattern Analysis, Risk Analytics, Policy Analysis, Policy Recommendations, Compliance Automation

Data & Analytics: SQL (PostgreSQL, MySQL, Oracle, Hive-compatible SQL), Python (Pandas, NumPy, scikit-learn, PySpark), Databricks (PySpark), Apache Spark, R, Statistical Analysis, Hypothesis Testing, Data Profiling, Feature Engineering, KPI Metrics (Precision, Recall, Chargeback Rate, Fraud Rate)

Visualization & Reporting: Tableau, Power BI, Matplotlib, Seaborn, Plotly, ggplot2, Quarto, Data Visualization, Dashboard Design

ML & Modeling: Random Forest, Logistic Regression, Isolation Forest, XGBoost, Statistical Modeling, Model Evaluation, scikit-learn, PyTorch, Hugging Face, NLP, BERT, spaCy

Engineering & Infrastructure: Python, FastAPI, REST APIs, AWS (S3, EC2, RDS, Glue), Docker, CI/CD (GitHub Actions), PostgreSQL, MySQL, ETL, Data Pipelines, Microservices, Libsodium, LangChain, LangGraph, RAG Pipelines, ChromaDB

Languages: Python (primary), SQL, R, PHP, JavaScript, TypeScript, PL/SQL, Bash